

Moroccan Darija Sentiment Analysis

The National School of Applied Sciences in Al Hoceima,
Abdelmalek Essaadi University

Authors: Asmae Daouad Kaoutar Asahaad

asmae.daouad@etu.uae.ac.ma kaoutar.asahaad@etu.uae.ac.ma

Supervision: Pr. Mohamed cherradi

Abstract

User opinions on products, services, and events are widely shared on social media, providing a valuable source of data for sentiment analysis. Analyzing comments and tweets written in Moroccan Darija helps identify public attitudes toward different topics. However, Darija poses several challenges due to its diglossic nature, the lack of standardized spelling, and the frequent use of Arabizi, which combines Latin characters and numbers. The main objective of this study is to compare three families of models for sentiment classification (Positive/Negative/Neutre) on a large corpus of Darija texts. Data were collected from multiple sources including online APIs and social media platforms, resulting in a large and diverse corpus of Darija comments. Three families of models were evaluated: classical Machine Learning models, Deep Learning models (LSTM, BiLSTM, CNN+LSTM), and Transformer-based models (DarijaBERT, AraBERT). The analysis revealed that Transformer models significantly outperform both classical and deep learning approaches. The best performing model was DarijaBERT-arabizi, achieving an accuracy of 90.05% and an F1-score of 90.04%. This result demonstrates that preserving the original Arabizi format without converting numbers to Arabic letters is more effective for Darija sentiment analysis, paving the way for improved NLP tools for non-standardized Arabic dialects.

Keywords— Sentiment analysis, Moroccan Arabic dialect, social media, Machine Learning, Deep Learning, Natural language processing.

1 Introduction

Natural Language Processing (NLP) is a key branch of artificial intelligence that enables computers to understand and process human language. This field tackles the complexities of language including variations in grammar, syntax, and semantics. Among its applications, sentiment analysis has garnered significant interest due to the rapid growth of user-generated content on digital platforms.

Sentiment analysis identifies and categorizes opinions expressed in textual data into categories such as positive, negative, or neutral. This task has become increasingly

important for organizations, businesses, and governments seeking to analyze public opinion and user feedback in real time, playing a crucial role in understanding social trends.

A major challenge for NLP lies in processing dialects, which often differ considerably from standard language. Moroccan Darija is a spoken Arabic vernacular with unique linguistic features including a diglossic nature, heavy borrowing from French, Spanish, and Amazigh, and no standardized written form. Its increasing presence in online communication makes it a challenging yet important target for NLP research.

This research leverages NLP and sentiment analysis to address the specific linguistic challenges posed by Darija, with the aim of developing improved sentiment analysis models for this dialect. This is particularly important given the growing use of Darija on social media, where understanding public sentiment can provide valuable insights for businesses, governments, and researchers.

2 Related Works

Several research efforts have explored various aspects of Natural Language Processing (NLP) for Moroccan Darija. While sentiment analysis has been a prominent area of focus, other NLP tasks have also received attention.

In the study by Sara Ouahab et al. [1], the authors constructed a large, multi-domain dataset of Moroccan Darija text collected from various social media platforms. This dataset enabled the development and evaluation of different NLP approaches, including transformer-based models such as BERT, highlighting the effectiveness of contextual representations for sentiment analysis in dialectal Arabic.

More recent research has focused on improving NLP performance for Moroccan Darija by addressing its linguistic complexity. Motassim Hamza [2] proposed an approach that combines Modern Standard Arabic (MSA) and Moroccan Darija for sentiment analysis. Their study emphasized the importance of handling code-switching phenomena, which are highly frequent in Moroccan social media and significantly affect NLP tasks such as sentiment classification.

Ouassil Mohamed Amine and Rabia Rachidi [3] analyzed sentiment classification on tweets written in Moroccan dialectal Arabic and Modern Standard Arabic. They evaluated multiple classifiers such as SVM, Logistic Regression, Random Forest, and LSTM using TF-IDF and n-gram features. Their results showed that SVM achieved competitive performance, reaching around 70% accuracy.

Abdeljalil Elouardighi et al. [4] proposed a machine learning-based sentiment analysis approach on Facebook comments written in both Modern Standard Arabic and Moroccan Darija. Their pipeline included preprocessing, stemming, feature extraction, and classification. The study demonstrated promising results and emphasized the importance of text normalization and feature engineering for dialectal Arabic.

Jbel Mouad et al. [5] focused on sentiment analysis of Moroccan dialect written in both Arabic and Latin scripts. They introduced a large dataset and performed experiments using several machine learning models. Their work highlighted the importance of handling script variation and showed that preprocessing significantly improves classification performance.

Amnay et al. [6] investigated the effectiveness of contextual embeddings for sentiment analysis in Moroccan Darija. They compared traditional embeddings (Word2Vec, FastText) with transformer-based models such as DarijaBERT. Their results showed that contextual models significantly outperform classical representations, especially when combined with neural classifiers like MLP, confirming the superiority of transformers for low-resource dialects.

Sakhi and Elfilali et al. [7] conducted a comparative study of machine learning models using the MAC dataset, which contains Moroccan Darija comments. They evaluated classifiers such as SVM, Naive Bayes, and KNN. Their findings highlighted that traditional machine learning approaches remain competitive but struggle with the linguistic variability of Darija compared to more advanced methods.

This research specifically contributes to the literature by focusing on sentiment analysis for Moroccan Darija. By addressing the unique challenges of nonstandardized dialects, this work aims to advance the broader field of sentiment analysis and enhance its real-world applicability.

3 Model Architecture

The proposed architecture for Darija sentiment analysis follows a sequential pipeline: Data Collection → Data Cleaning → Data Processing → Feature Extraction → Sentiment Classification.

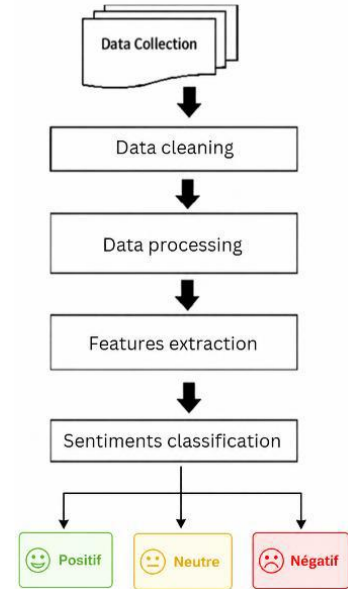


Figure 1: Proposed Architecture for Darija Sentiment Analysis.

4 Methodology

This chapter details the methodology employed for classifying sentence sentiment. It describes the text models, dataset, and classifiers used, along with the preprocessing and normalization techniques designed to handle the informal nature of dialectal Arabic, particularly Moroccan Darija. The evaluation metrics used to assess sentiment classification performance are also presented. The methodology can be summarized as follows: a description of the dataset used in this work; the application of various preprocessing steps, including normalization, noise removal, and conversion of emoticons to text, to enhance classification performance; an explanation of the techniques used for text representation; and finally, the classification of Moroccan dialect text using machine learning, deep learning, and finetuned pretrained model classifiers, followed by a comparison of the results.

4.1 Data Collection

The data collection process involved multiple sources to ensure diversity in the dataset:

a-Existing Datasets: Five different pre-existing datasets were leveraged:

-Dataset 1 MAC (Moroccan Arabic Corpus) [8]: A free and large corpus consisting of 18,000 manually labeled tweets, MAC is the first and largest Moroccan Arabic corpus for sentiment analysis, praised for its size, and accessibility to

the research community. It provides a benchmark for future work, including polarity classification and language identification.

-Dataset 2 OMCD (Offensive Moroccan Comments Dataset) [9]: This dataset focuses on detecting offensive content, such as verbal attacks and hate speech, in Moroccan Arabic on social media. It is the first of its kind for the Moroccan dialect, offering annotated data for detecting offensive language in Dialectal Arabic (DA).

-Dataset 4 ElecMorocco2016 [10]: A sentiment analysis dataset containing 10,254 Arabic Facebook comments related to the 2016 Moroccan elections. The comments are written in both Standard Arabic and Moroccan Dialect (Darija), offering insights into political sentiments during the election period.

- Dataset 5 Darija Reviews Dataset [11]: A sentiment analysis dataset collected from Hugging Face containing user reviews written in Moroccan Darija. The dataset includes informal Arabic dialect text used in real-world contexts, making it suitable for sentiment analysis tasks in Moroccan dialect. It provides valuable linguistic variations of Darija expressions used in online platforms.

b-YouTube/Instagram Data Scrapping: a customized data collection process was conducted using web scraping of social media platforms, primarily **YouTube** and **Instagram**, where Moroccan users frequently express opinions and emotions using Darija written in both Arabic and Latin scripts.

To facilitate the tagging process, a specific scraping strategy was adopted based on the nature of the collected content. Positive samples were extracted mainly from entertaining, humorous, motivational, and emotionally positive videos, associated with supportive and favorable comments. Negative samples were collected from videos related to sad, dramatic, or controversial topics that generally elicit negative emotional reactions.

The collected data were manually reviewed and organized into categories of positive, negative sentiment. This approach allowed for the construction of a diverse dataset that reflects actual online communication in Morocco and the linguistic complexity of Moroccan Darija.

4.2 Data Cleaning and Preprocessing

Cleaning and preprocessing techniques are fundamental in Sentiment Analysis (SA) for Arabic dialectal text, particularly for the Moroccan Darija dialect. This necessity arises from the unique challenges presented by social media text, which is characterized by its informal nature and lack of standardization. The implementation addresses these challenges through a comprehensive preprocessing pipeline specifically designed for Darija text, operating at multiple levels of text normalization.

Data Cleaning :

Cleaning is focused on eliminating irrelevant or noisy elements from the dataset. This step ensures that the data is free from non-informative parts, which could negatively affect the model's performance. The cleaning steps include:

Noise Removal: Removal of special characters, punctuation marks, numbers in both Arabic and Latin scripts, redundant or spamming alphabets, and other unwanted symbols that do not contribute to the sentiment analysis task.

Context-Aware Cleaning: Handling modern social media elements, including:

- **Emojis translation** to text .
- Removal of URLs, Emails, underscores, special tags (e.g., & and ;), and HTML tags.
- **Stop Words Removal** is an important preprocessing step in sentiment analysis [12].

Moroccan dialect stop words lists	
Stop words scripted in Arabic letters	Stop words scripted in Latin letters
['ila','3a','ta','idan','9al','chwiya','ktar','bzafl', 'chi',...]	['أقل','أحيث','أنتا','أمعانا','أطى', 'ألا','أكثر',...]

Figure 1: Examples of Moroccan dialect stop words.

Data Preprocessing :

Preprocessing techniques prepare the text for effective analysis by transforming it into a standard format. This step includes:

Arabic Text Normalization: is an important preprocessing step in sentiment analysis. It includes the removal of diacritical marks (**tashkeel**) and elongation characters (**tatweel**) to reduce text variability. For example in Darija written Arabic we

can see forms like $\text{مزيان} \rightarrow \text{مزيان}$, and $\text{كبير} \rightarrow \text{كبير}$ where elongation is removed to normalize the word. The process also standardizes different forms of Hamza and Alef into a unified format such as $\text{اسمي} \rightarrow \text{اسمي}$, $\text{أمل} \rightarrow \text{أمل}$, $\text{إمتى} \rightarrow \text{إمتى}$. In addition, it removes the definite article “ال” prefix when necessary, for example $\text{الدار} \rightarrow \text{دار}$, $\text{المدينة} \rightarrow \text{مدينة}$ helping produce cleaner and more consistent textual data for analysis in Darija sentiment analysis tasks.

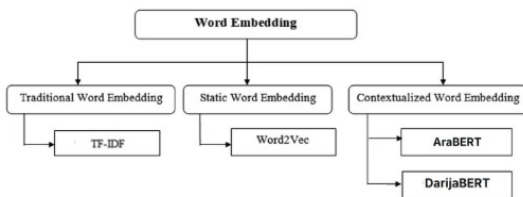
UnNormalized Arabic Letters	Normalized Letters
Hamza(ء, ؤ)	ء
Alif(ا, آ, إ, ؤ)	ا
Alif Lam(ال, آل)	ال
Hae(ة)	ه
Dad(ظ, ض)	ض

Arabizi Normalization : During preprocessing, we use a mapping dictionary to convert common Arabizi numbers into their Arabic letter equivalents, such as $7 \rightarrow \text{ح}$ and $3 \rightarrow \text{ع}$. This normalization step helps clean and standardize text data, improving sentiment analysis performance.

$7 \rightarrow \text{ح}$
 $3 \rightarrow \text{ع}$
 $9 \rightarrow \text{ق}$
 $2 \rightarrow \text{أ}$
 $5 \rightarrow \text{خ}$

3.1 Feature Extraction

Feature extraction is a key step in converting textual data into numerical representations suitable for analysis. In this study, several methods were explored, including TF-IDF, Word2Vec, AraBERT, and DarijaBERT.



TF-IDF [13]: A classical statistical method that encodes text based on the frequency and relative

importance of words cross the corpus. Used with classical machine learning classifiers, TF-IDF is effective for capturing discriminative vocabulary but does not encode semantic relationships.

Word2Vec [14]: A neural embedding method that learns dense vector representations capturing semantic similarity between words, used as input features for deep learning models.

AraBERT [15]: A pre-trained BERT-based model for Ara-bic, providing contextualized word embeddings that encode both semantic and syntactic relationships, fine-tuned on our dataset.

DarijaBERT [16]: A BERT-based model specifically pre-trained on Moroccan Darija, capturing unique vocabulary and code-switching patterns more effectively than general Arabic models.

5 Classification Models :

Naive Bayes (NB) [17]: Naive Bayes is a probabilistic machine learning classifier based on Bayes’ theorem. It assumes conditional independence among features, making it computationally efficient and particularly suitable for text classification tasks. Due to its simplicity and effectiveness, it is commonly used as a baseline model in sentiment analysis.

Support Vector Machine (SVM)[18]: Support Vector Machine is a supervised learning algorithm that identifies the optimal hyperplane separating different classes in a high-dimensional feature space. When combined with TF-IDF features, SVM demonstrates strong performance in text classification and sentiment analysis tasks.

BiLSTM [20]: Bidirectional Long Short-Term Memory (BiLSTM) extends the LSTM architecture by processing text sequences in both forward and backward directions. This approach enables the model to capture contextual information from both past and future words, leading to a richer representation of textual data.

CNN [21]: Convolutional Neural Network (CNN) is a deep learning architecture that applies convolutional filters to extract local features and n-gram patterns from text representations. By identifying the most relevant patterns through convolution and pooling operations, CNN effectively captures important semantic and syntactic information, making it a powerful model for text classification and sentiment analysis tasks.

AraBERT [15]: and DarijaBERT [16]: AraBERT and DarijaBERT are Transformer-based language models that rely on self-attention mechanisms to process text sequences and capture contextual relationships. After fine-tuning on labeled sentiment datasets, they effectively model semantic, contextual, and dialect-specific characteristics, making them highly suitable

for sentiment analysis of Arabic and Moroccan Darija texts.

6 Results and discussion :

6.1 Comparative study of created models

In this section, we present a **comparative** experimental study of different models developed for Moroccan sentiment analysis. The objective is to evaluate and identify the most effective approach for sentiment classification on Moroccan Darija text. The experiments include **both classical machine learning** models and **deep learning architectures**, as well as **pre-trained transformer-based** models fine-tuned on our dataset.

The performance of all models is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score.

Model	Prec.	Rec.	F1	Acc.
Naive Bayes	0.78	0.79	0.79	0.79
SVM	0.84	0.84	0.84	0.84
BiLSTM	0.86	0.86	0.86	0.86
CNN	0.83	0.86	0.84	0.86
AraBERT	0.86	0.87	0.87	0.86
DarijaBERT	0.88	0.92	0.90	0.90

Table 1: Comparative performance of all models on Moroccan Darija sentiment classification.

The experimental results presented in the comparison table allow a clear evaluation of the performance of the different models. Among the classical machine learning classifiers, Support Vector Machine (SVM) achieved the best performance with an accuracy of 84.14%, followed by Naive Bayes with 79.12%.

In contrast, deep learning models show a noticeable improvement over classical approaches. The BiLSTM model achieved the highest performance among neural architectures with an accuracy of 86.40%, demonstrating its ability to capture sequential dependencies in the text. Other recurrent-based models also show competitive results, confirming the effectiveness of deep learning for sentiment classification tasks.

The best results are obtained by transformer-based models, which significantly outperform both classical and deep learning approaches. DarijaBERT-arabizi achieves the highest overall performance with an accuracy of 90.05% and an F1-score of 90.04%. It is followed by DarijaBERT with 88.28% accuracy, while AraBERT achieves 86.87%. These

results highlight the advantage of domain-specific pre-trained models, particularly when trained on Moroccan dialect data.

Finally, **DarijaBERT-arabizi** achieves the best overall performance among all tested models, making it the most effective approach for Moroccan Darija sentiment analysis.

7 Comparison with Related Work :

To further validate our results, we compare our best-performing model with the results reported by Jbel et al. [5], who evaluated classical and deep learning models on Moroccan Darija sentiment analysis. As shown in Table 2, their best model (CNN) achieved an accuracy of 85% and an F-measure of 85%. In contrast, our DarijaBERT model achieves a significantly higher accuracy of 90.05% and an F1-score of 90.04%, demonstrating a clear improvement of approximately **5 percentage points**. This confirms that Transformer-based models, particularly those pre-trained on Moroccan Darija data, substantially outperform classical machine learning and standard deep learning approaches for dialectal Arabic sentiment analysis.

Model	Acc.(%)	Prec.(%)	Rec.(%)	F1(%)
<i>Jbel et al. [7]</i>				
SVM	73.5	72.9	73.0	72.8
NB	71.5	71.0	71.0	70.8
KNN	68.6	68.0	69.0	69.0
CNN	85.0	84.0	84.0	85.0
LSTM	84.0	83.7	84.0	83.5
<i>Our Work</i>				
DarijaBERT	90.05	88.0	92.0	90.04

Table 2: Comparison with Jbel et al. [5].

8 Limitations of the Approach

Although Moroccan Darija is increasingly used in digital communication platforms, its automatic processing remains challenging due to several linguistic and sociolinguistic factors. These characteristics introduce complexity when applying Natural Language Processing (NLP) techniques.

• Regional Variations of Darija :

Our dataset, while diverse, may not fully represent all **regional variations of Darija**. Morocco has **multiple regions (Oujda, Casablanca, Fès, Tanger, the South, etc.)**, and each region has its own specific vocabulary, expressions, and even different ways of writing the same word. For example, a word that is positive in Casablanca might be neutral or even negative in Oujda. These regional variations are not fully captured in our current dataset. Future work

should collect more data from different Moroccan regions to build a more representative model.

• **Absence of a Standard Writing System:**

Moroccan Darija does not follow a fixed orthographic standard, which leads to high variability in spelling across users. The same word can appear in multiple forms, for example:
 – “good” may be written as *mzian*, *mezyan*, or *meziane*
 – “what” may appear as *chno*, *chnou*, *chnowa*, or *cheni*

• **Code-Mixing-Phenomenon:**

Darija is often mixed with other languages such as French, English, and Modern Standard Arabic within the same sentence. This code-switching increases the complexity of linguistic processing. For instance:
 – *khas had weekend na9ra deeplearning o nkhdem project*.
 – *C'est bon, asahbti*

9 Conclusion

In this work, we explored different approaches for Moroccan Darija sentiment analysis using classical, deep learning, and transformer-based models. The experimental results show that transformer-based approaches provide the best overall performance, particularly models adapted to Moroccan Darija text. The study also highlights the importance of preprocessing and suitable text representation techniques for improving sentiment classification accuracy.

Despite the promising results, Moroccan Darija remains a challenging language for NLP tasks due to its informal nature, spelling variations, and frequent code-switching. Nevertheless, the obtained results demonstrate the potential of deep learning and pre-trained language models for handling dialectal Arabic sentiment analysis effectively.

Looking ahead, we plan to further improve our sentiment analysis system by extending it beyond binary classification to support additional sentiment categories such as anger, happiness, sadness, and surprise. We also aim to collect larger and more diverse datasets covering different Moroccan regional accents and variations of Darija in order to build more robust and generalized models.

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